

1. Maintenance — Mechanical Seal Leakage

Title:

Mechanical Seal Leakage

Type:

MAINTENANCE

Subject Key:

p-101/seal

Date:

2026-07-14

Content:

During routine inspection of centrifugal pump P-101, a small amount of process fluid was observed around the mechanical seal housing. The pump remained operational and no abnormal vibration was recorded during this inspection. Seal leakage was classified as minor and the area was cleaned for continued monitoring. Technician recommended checking seal condition and shaft alignment during the next maintenance window.

2. Observation — Increased Vibration

Title:

Increased Pump Vibration

Type:

OBSERVATION

Subject Key:

p-101/vibration

Date:

2026-07-21

Content:

Operator reported intermittent rattling from pump P-101 during operation at normal process load. Vibration appeared higher than the previous inspection, particularly during startup. Suction pressure was recorded at approximately 2.1 bar. No visible external damage was found. Further inspection of suction conditions, bearings, coupling alignment, and possible cavitation was recommended.

3. Maintenance — Seal Replacement

Title:

Mechanical Seal Replacement

Type:

MAINTENANCE

Subject Key:

p-101/seal

Date:

2026-07-28

Content:

The mechanical seal on pump P-101 was replaced following continued leakage. The replacement was completed during a scheduled maintenance window. Shaft condition and coupling alignment were checked before reassembly. The pump was returned to service after a leak check. No immediate leakage was observed after startup.

4. Incident — Repeated Seal Failure

Title:

Repeated Mechanical Seal Failure

Type:

INCIDENT

Subject Key:

p-101/incident

Date:

2026-08-02

Content:

Pump P-101 experienced renewed mechanical seal leakage approximately one week after the previous seal replacement. The recurrence was recorded as INC-1042. Initial investigation identified unstable suction conditions during part of the operating period. The cause was not conclusively established. Recommended follow-up includes checking suction restrictions, operating conditions, shaft alignment, bearing condition, and seal installation.

5. Procedure — Mechanical Seal Inspection

Title:

P-101 Mechanical Seal Inspection Procedure

Type:

PROCEDURE

Subject Key:

p-101/procedure

Date:

2026-08-05

Content:

Before replacing the mechanical seal on P-101, technicians should inspect the seal housing for leakage patterns and contamination. Check shaft condition, shaft runout, coupling alignment, bearing condition, seal faces, elastomers, and installation seating. Verify that process conditions are within the pump operating range. Inspect suction pressure and suction-side restrictions before concluding that the seal itself is the primary cause of failure.

6. Observation — Suspected Cavitation

Title:

Possible Cavitation During Operation

Type:

OBSERVATION

Subject Key:

p-101/cavitation

Date:

2026-08-18

Content:

During inspection of pump P-101, intermittent rattling and elevated vibration were again observed. The symptoms were more noticeable when the pump was operating near a higher flow condition. Suction pressure remained approximately 2.1 bar during the observation. Cavitation was identified as a possible contributing condition, but the available evidence is insufficient to confirm cavitation as the root cause.

7. Maintenance — Repeated Seal Failure Investigation

Title:

Seal Failure Investigation

Type:

MAINTENANCE

Subject Key:

p-101/investigation

Date:

2026-08-28

Content:

A follow-up investigation was performed after repeated mechanical seal leakage on P-101. The maintenance team reviewed the previous seal replacement, operating observations, vibration reports, and suction pressure readings. Evidence indicates repeated seal failures occurred alongside intermittent vibration and unstable suction conditions. Shaft alignment and bearing condition were inspected. The investigation recommends verifying suction-side conditions and operating range before another seal replacement. Root cause remains unconfirmed.

8. Procedure — Cavitation Investigation

Title:

P-101 Cavitation Investigation Procedure

Type:

PROCEDURE

Subject Key:

p-101/cavitation-procedure

Date:

2026-08-30

Content:

For suspected cavitation on P-101, technicians should first verify suction pressure and suction-side restrictions. Check whether the pump is operating within its intended flow range. Inspect the suction line, valves, filters, and upstream conditions. Compare vibration and noise observations across operating conditions. Inspect impeller condition if

operational checks indicate persistent cavitation. Cavitation should not be confirmed solely from rattling or vibration; operating and inspection evidence should be reviewed together.

Questions to manually test

These 8 records give you a much better test matrix.

Test 1 — Basic retrieval

Why does P-101 keep experiencing mechanical seal failures?

Expected evidence should include records around:

- Mechanical Seal Leakage
- Mechanical Seal Replacement
- Repeated Mechanical Seal Failure
- Seal Failure Investigation

The answer should **not claim a confirmed root cause**.

Test 2 — Procedure retrieval

What should a technician inspect before replacing the seal again?

Expected sources:

- P-101 Mechanical Seal Inspection Procedure
 - Seal Failure Investigation
 - Repeated Mechanical Seal Failure
-

Test 3 — Cavitation

What evidence suggests possible cavitation on P-101?

Expected sources:

- Possible Cavitation During Operation
- Increased Pump Vibration
- Cavitation Investigation Procedure

The answer should distinguish **possible cavitation** from **confirmed cavitation**.

Test 4 — Exact identifier

| What incident was associated with the repeated seal failure?

Expected:

INC - 1042

This tests your keyword/identifier retrieval.

Test 5 — Timeline

| What happened to P-101 between July 14 and August 28?

This tests whether the model can synthesize the chronological records.

Test 6 — Evidence limitation

Ask:

| What is the confirmed root cause of the P-101 seal failures?

A good answer should **not invent one**.

It should say that the available records identify possible contributing conditions, including unstable suction and possible cavitation, but the root cause remains unconfirmed.

This is an important demo because it shows **grounded AI rather than hallucinated diagnosis**.

Test 7 — Exact measurement

| What suction pressure was recorded during the P-101 observations?

Expected:

Approximately 2.1 bar